Dynamic Divided Attention across Spatial Locations

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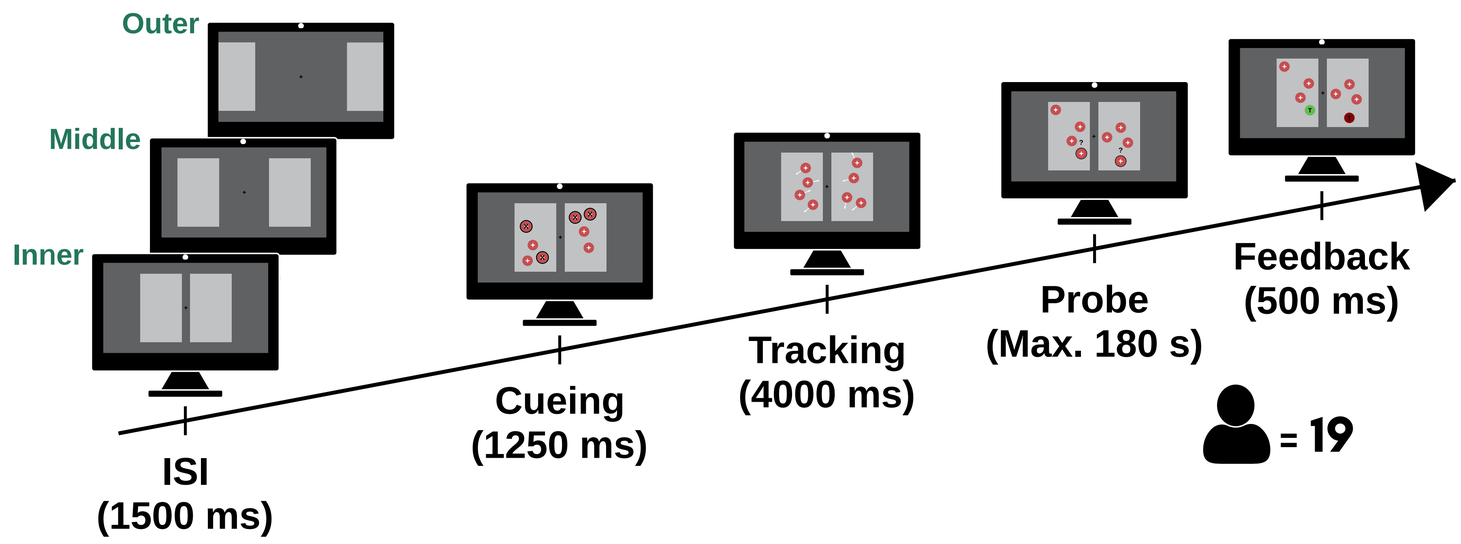
Introduction

The ability to attentionally track moving objects is essential to many everyday situations such as driving, team sports, or social interactions in groups. Therefore, it is little wonder, that we're abled to across the broad visual field (Franconeri et al., 2008) and scales down to areas, minuscule in size (Cavanagh & Alvarez, 2005).

Yet, it remains unknown how **distance** from *central fixation point* **affects** tracking **performance** and **attentional allocation**.

Thus, we ask: "How does one's ability to **divide attention** between dynamic objects change with the spatial location of the objects to track?"

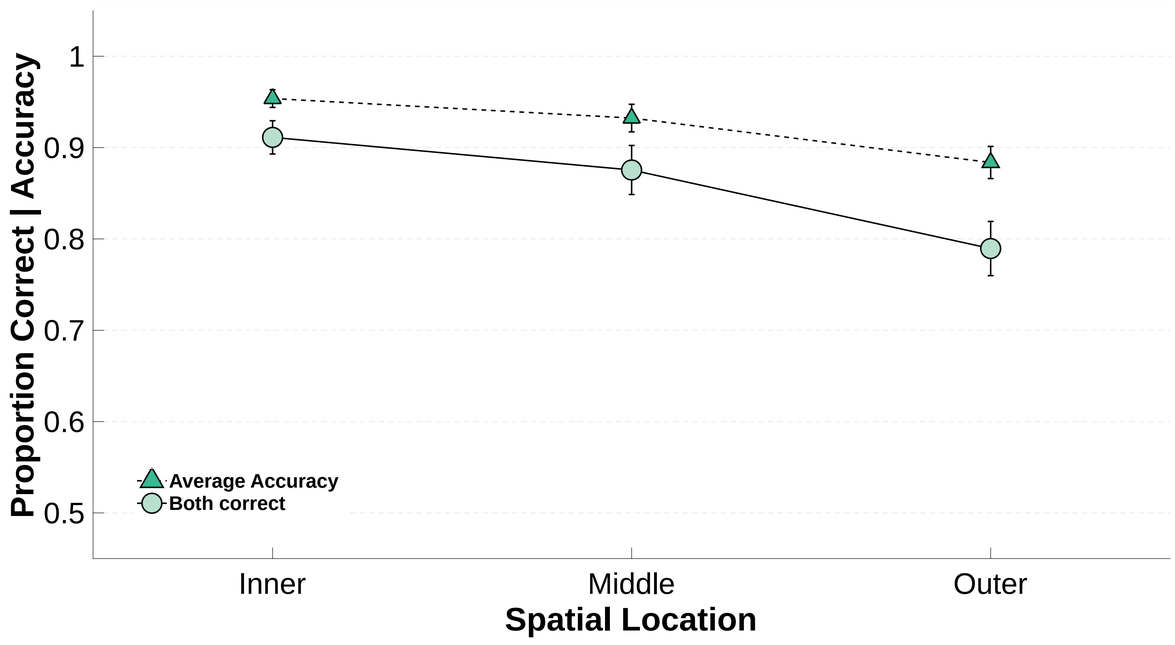
Methods



Participants **bilaterally** tracked four marked objects out of a total of eight, while maintaining **central fixation** (Multiple Object Tracking).

Concurrent measurement of divided attention in visual cortex through the EEG recording of frequency tagged **steady-state visual evoked potentials** (Regan, 1966).

Results | Behavioral

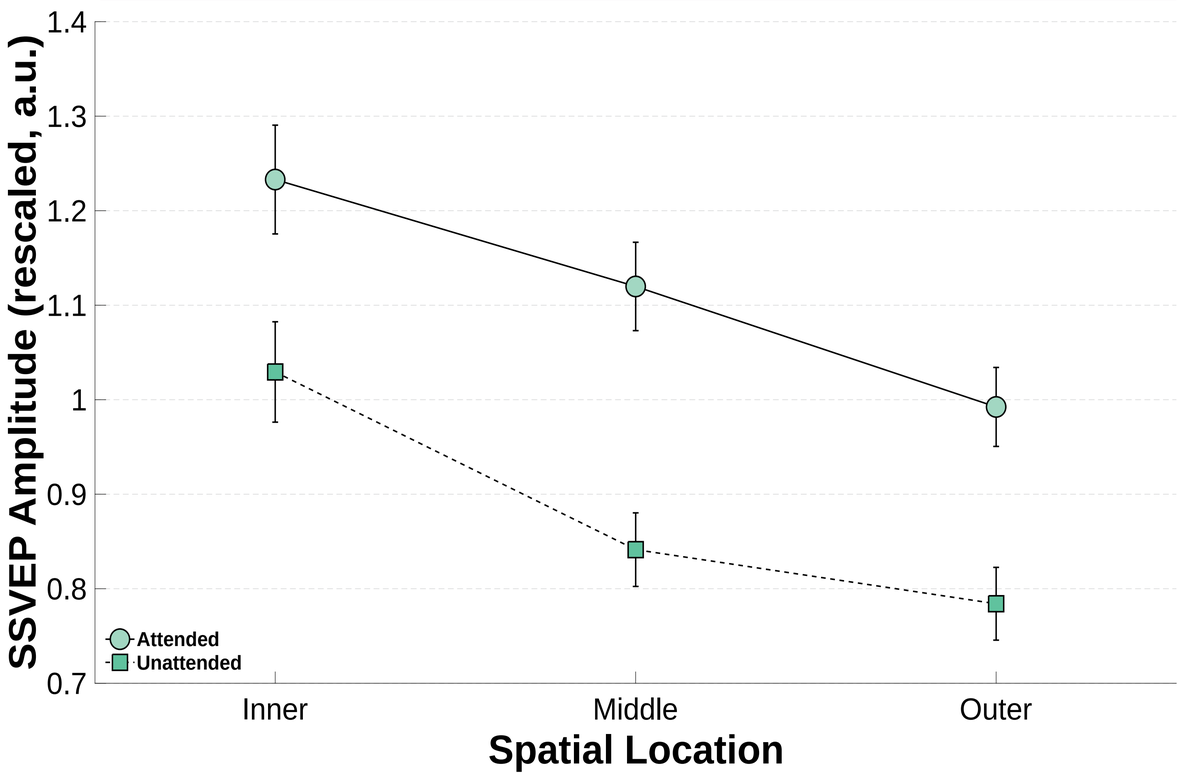


One-way repeated measures ANOVA | Average Accuracy

	F(2, 36)	p (GG)	Part. ω^2
Spatial Location	31.60	$< 10^{-6}$	0.611

Tracking performance **decreases** with **increasing distance** from fixation.

Results | EEG



Repeated-Measures ANOVA | Attention × Spatial Location

Effect	F (df)	p (GG)	Part. ω^2
Spatial Location	6.39 (2, 36)	0.0119	0.216
Attention	43.26 (1, 18)	$< 10^{-5}$	0.679
Spatial Loc. × Att.	0.97 (2, 36)	0.383	0.000

Note: GG = Greenhouse-Geisser correction

For increasing distance to the fixation point, behavioral decrease is also reflected in amplitude, while **attentional allocation** seems to **remain constant**.

Overall Conclusions

Divided Attention in terms of Spatial Locations depends on the **difficulty** of a task, therefore, **position** in **space**, **relative** our **point of fixation**.

There is no impairment to **attentional allocation**. Indeed, the evidence suggests that its expression remains **unaffected** by a **decline** in **behavioural performance**.

The findings of this study suggest that **Divided Attention** may act as a **protective factor** for **behavioural performance**, with its power being **contingent** to **task complexity**.

References

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